# Can Jac access order guide GPU memory layout?

Primary variable: node layout • same checked LLVM O2 kernel for both Jac arms

BEFORE: Jac Random928 + checked LLVM O2

AFTER: Jac Predicted + identical checked LLVM O2 PTX

REFERENCE: fixed handwritten tiled CUDA, nvcc O3

$256^3$: 947.380 → 540.418 µs; layout speedup 1.753×.

Predicted / handwritten: 26.10×. Full output/status validation on a shared RTX 3090.

# Place nodes in their known discovery order

C[i,j] = Σq A[i,q] × B[q,j] • one Jac GPU lane per output

Predicted concatenates complete cursor-a BFS and complete cursor-b BFS: all A nodes, then all B nodes.

| Physical ID order | Logical node sequence |
|---|---|
| Random928 | Fixed seeded permutation of all A and B identities |
| Predicted: cursor a | A[0,0], A[1,0], … ; A[0,1], A[1,1], … |
| Predicted: cursor b | B[0,0], B[0,1], … ; B[1,0], B[1,1], … |

Values, tags, CSR rows/targets, heads and initialized queues are remapped consistently. Saved bijections and every q/cursor/lane arrival preserve logical identity.

Jac source, arithmetic, queue/visit limits, lane mapping and grid/block stay fixed.

# A matched optimizing policy and resident metric

LLVM O2 and nvcc O3 are distinct optimizing pipelines with different guarantees.

Jac: same production raw LLVM → standard checked LLVM O2 → existing target codegen. Both layouts reuse one loaded function.

CUDA: fixed 16 × 16 shared-memory tiled int64 kernel, nvcc -O3. The same kernel handles all five sizes.

One randomized interleaved run: 3 warmup rounds, then 15 rounds × 16 graph nodes per cell. Actual Nsight node-kernel end − start is the primary metric.

Packing, transfers, poisoning and verification are separate from resident kernel time. Shared RTX 3090; Ollama held approximately 23,462 MiB.

# Three primary groups, full matmul coverage

Median actual kernel duration (µs), 240 samples per cell; lower is faster.

| M × K × N | Jac BEFORE Random + O2 | Jac AFTER Predicted + O2 | REFERENCE Tiled CUDA O3 |
|---|---|---|---|
| 3 × 5 × 2 | 6.272 | 7.232 | 1.568 |
| 17 × 31 × 19 | 42.304 | 45.152 | 2.208 |
| 64 × 128 × 64 | 200.929 | 180.865 | 5.440 |
| 128 × 128 × 128 | 187.265 | 164.032 | 4.832 |
| 256 × 256 × 256 | 947.380 | 540.418 | 20.705 |

All three columns share one fresh randomized interleaved trace: optimized-profile.

Jac uses 128 threads/block. CUDA uses its fixed 16 × 16 tile geometry.

# Layout benefit depends on workload size

Before / after > 1 means layout improves Jac; after / reference is the residual gap.

| M × K × N | Layout speedup Random / Predicted | Residual gap Predicted / CUDA |
|---|---|---|
| 3 × 5 × 2 | 0.867× | 4.61× |
| 17 × 31 × 19 | 0.937× | 20.45× |
| 64 × 128 × 64 | 1.111× | 33.25× |
| 128 × 128 × 128 | 1.142× | 33.95× |
| 256 × 256 × 256 | 1.753× | 26.10× |

Seeds 928, 929 and 930 agree on direction at each size: regression on the two smallest cases, improvement on the three larger cases.

Raw samples, p10/p90 and paired-round means are retained. The separate ordinary-event run gives the same directions.

# Address organization is directly reproducible

Deterministic logical address analysis • $256^3$ • mean 32-byte segments per warp access

| Logical access | Random928 | Predicted |
|---|---|---|
| Cursor a: value | 1.000 | 1.000 |
| Cursor b: value | 31.987 | 8.000 |
| Cursor b: csr_begin | 31.985 | 9.000 |
| Cursor b: csr_target | 31.969 | 8.000 |

At N=256, lanes share one A node and access 32 distinct B nodes. Predicted makes those B-node addresses consecutive.

Nsight Compute returned ERR_NVGPUCTRPERM. Segment counts model logical addresses; hardware transactions, cache misses and occupancy remain unmeasured. Hardware attribution stays unresolved.

# Correctness and launch evidence

Byte-identical checked O2 artifact for both Jac layouts; every replay checked

| Evidence | Result |
| --- | --- |
| Logical graph and traversal equivalence | All fields, edges, heads, permutations and lane arrivals pass |
| Full normal-case outputs/statuses | Poison → replay → check all M×N values and Jac statuses |
| Prior error/status validation, identical PTX | 13 O2 fixtures match production host JIT |
| Independent Nsight launch join | 7,225 launches; 6,000 measured; 3,600 primary nodes |

Error fixtures cover overflow, invalid node/CSR, queue capacity and round limits. Fresh output checks use NumPy int64 plus the saved oracle.

Both optimizing pipelines retain their execution contracts. The remaining CUDA gap compares complete implementations.

# $3000^3$: separate correctness, timing and capacity

Historical evidence uses its own older PTX/data and 32,768-output timed subsets.

Historical full correctness: 9,000,000 outputs. Historical resident subset medians: Random 70.802 ms, Predicted 8.973 ms (7.89×); handwritten timing absent.

One packed graph: 719.95 MB. This run: 357.83 MB free after context creation.

A paired full-coverage follow-up needs about 1.66 GB plus allocation/JIT headroom, then all 275 bounded Jac batches and matched tiled output coverage.

Evidence: scripts/gpu_walker/layout_study/validation/20261022/REPORT.md and comparison.csv. Reproduce with layout_study/reproduce.sh; deck source hashes are in sources.json.